

Lesson 1 · Practice sheet

NCEA Level 2 · AS 91262 · Gradients, differentiation and tangents

Time: 30 minutes

Answer version · tutor's working, not an official mark scheme

Show all working. Write $f'(x)$ every time you use calculus. The letter at the right shows the level: [A] Achieved, [M] Merit, [E] Excellence.

1. Differentiate $f(x) = 6x^4 - 3x^3 + 2x - 11$. [A]

$$f'(x) = 24x^3 - 9x^2 + 2$$

2. Find the gradient of $y = x^3 - 4x^2 + 5$ at the point where $x = -1$. [A]

$$\frac{dy}{dx} = 3x^2 - 8x$$

$$\text{At } x = -1: 3(-1)^2 - 8(-1) = 3 + 8 = 11, \text{ so the gradient is } 11$$

3. Find the coordinates of the point on $f(x) = 3x^2 - 12x + 2$ where the gradient is 6. [A]

$$f'(x) = 6x - 12 = 6, \text{ so } x = 3$$

$$f(3) = 27 - 36 + 2 = -7$$

$$\text{The point is } (3, -7)$$

4. Find the equation of the tangent to $y = x^3 - 3x^2 + 2$ at $x = 3$. [A]

$$\text{Point: } y = 27 - 27 + 2 = 2, \text{ so } (3, 2)$$

$$\text{Gradient: } \frac{dy}{dx} = 3x^2 - 6x, \text{ so } m = 27 - 18 = 9$$

$$y - 2 = 9(x - 3)$$

$$y = 9x - 25$$

Question 5 [A]

- (a) A function f is given by: $f(x) = 4x^3 - 2x^2 - 7x + 4$

Use calculus to find the gradient of the graph of the function at the point where $x = 3$

$$f'(x) = 12x^2 - 4x - 7$$

$$f'(3) = 12(3)^2 - 4(3) - 7 = 108 - 12 - 7$$

$$\text{gradient} = 89$$

Question 6 [A]

- (b) For the cubic function $f(x) = \frac{1}{2}x^3 + \frac{1}{2}x$ find the equation of the tangent to the curve at $x = 2$

$$\text{Point: } f(2) = 0.5(8) + 0.5(2) = 5, \text{ so } (2, 5)$$

$$f'(x) = 1.5x^2 + 0.5, \text{ so } m = f'(2) = 1.5(4) + 0.5 = 6.5$$

$$y - 5 = 6.5(x - 2)$$

$$y = 6.5x - 8$$

Question 7 [A] · Merit if all steps are linked

- (b) The function f is given by: $f(x) = 5 + 3x + cx^2 - 2x^3$

At the point on the graph of the function where $x = 2$, the gradient is -5 .

Find the value of c .

$$f'(x) = 3 + 2cx - 6x^2$$

$$f'(2) = 3 + 4c - 24 = -5$$

$$4c = 16, \text{ so } c = 4$$

Question 8 [A]

- (a) A function f is given by $f(x) = 2x^4 + 4x^3 - 20x^2 - 5$.

Use calculus to find the gradient of the graph of the function at the point where $x = 3$.

$$f'(x) = 8x^3 + 12x^2 - 40x$$

$$f'(3) = 8(27) + 12(9) - 40(3) = 216 + 108 - 120$$

$$\text{gradient} = 204$$

Question 9 [M]

- (d) The tangent to the curve $f(x) = px - qx^2$ at the point $(2, -10)$ has a gradient of -6 .

Find the values of the constants p and q .

On the curve: $f(2) = 2p - 4q = -10$

Gradient: $f'(x) = p - 2qx$, so $f'(2) = p - 4q = -6$

Subtract: $p = -4$; then $-4 - 4q = -6$, so $q = 0.5$

$p = -4, q = 0.5$

Question 10 [A]

- (a) Find the equation of the tangent of the curve $y = 2x(x - 3)$, where $x = 1$.

Expand: $y = 2x^2 - 6x$; at $x = 1, y = 2 - 6 = -4$, point $(1, -4)$

$y' = 4x - 6$, so $m = 4(1) - 6 = -2$

$y + 4 = -2(x - 1)$, so $y = -2x - 2$